

The memory reflex

**Why experienced time presupposes memory rather than arising
from it (Doc. 304)**

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Memory kernel: M_τ and the bounded kernel are assigned to the case of *constant* defect ($d = 1/75$ fixed, linear growth $D(10^5) = 1333$), not to the case $d = 0$ (there $D(k) \equiv 0$, no accumulation); previously “degenerate $d = 0$ / Markov limit”. Carried through likewise in the sections on the measurement level and on curvature fidelity.

Doc. 304 The memory reflex — why experienced time presupposes memory rather than arising from it

Abstract. A common construction makes experienced time the *product* of memory. This document reverses the order: memory presupposes time rather than producing it — the *memory reflex*, counterpart of the word reflex of Doc. 301. In FFGFT time is fundamental, the cumulated defect $D(k) = \sum 100 \xi_n \rightarrow \ln(1 + k/75)$ of the unrolled winding; and the difference between a finite, forgetting kernel and an accumulating structure is already a fact about time, not first about memory. FFGFT itself does not need that split — it carries a single, curvature-faithful winding —; the split appears only where the time structure is projected onto a cognitive model. On the measurement side a second reflex follows: a measurement excerpt

is reference-less, with no absolute place on the spiral, and the apparent uniformity of time is a *linearisation artefact* with a verified error quantity $e = x - \ln(1 + x)$ and the scale-invariant constant $1 - \ln 2$; this linearisation sits in the *projection* of the result, not in the measurement. Rolled up, the 75 parts are exactly congruent; unrolled, only asymptotically self-similar; what survives the loss of reference is the ordering alone. In the FFGFT reading it is *projection-fidelity*, not reach, that discriminates — offered as an additional, rejectable bridge candidate (restricted, Category B), not as a replacement of any external memory definition, in a symmetric relationship between the frameworks. Inference from measurement to cause succeeds only up to a holographic blur: presence is not reconstructibility. Booking is kept separate throughout — mathematically derived, bridge (restricted), posit and reflex diagnoses (P35).

A common construction explains experienced time as the *product* of memory: physical evolution → physical records → recursive memory → self-related representation → experienced time. In this order, time is the end product and memory its precursor. This document shows that the order belongs the other way round. Memory without time does not function — it already presupposes time. The impression that time arises from memory is the same inversion as “it from bit”: the memory reflex, the counterpart of the word reflex of Doc. 301, one level down. The same inversion appears a second time on the *measurement* side — as the linearization reflex —, this time with a verified, scale-invariant error quantity; and a third time in the inference from part to whole — as the extrapolation reflex. This document treats all three.

.1 The narrowing and its price

The original ambition was the full temporal triad: causal ordering (U1), present phase (U2), and memory / non-Markovian history (U3)¹, joined into a past–present–future world-model. Over time the discussion narrowed to U3 alone, then to the boundary between a finite memory kernel M_τ and an accumulative structure M_B , and finally to an operational benchmark question. This narrowing is methodologically legitimate — only the isolable, falsifiable piece can be tested — but it has a price: U1 and U2 drop out of the test, and the phenomenon itself was their *interplay*, not the component. Even a passed memory test would show “semantically structured memory is distinguishable from generic compression”, not “here is experienced time”.

¹The temporal sectors U1 (causal ordering / forward updating), U2 (phase and coherence) and U3 (memory, path dependence, recursive access), together with the notation M_τ/M_B , are adopted from Krüger's Spiral-Time Memory work (2026); there M_τ denotes finite, weighted, or compressible physical memory and path dependence, M_B a persistent biological-cognitive architecture (recursive access, semantic organisation, affective weighting, identity binding, persistent state carry, predictive reuse), defined operationally and functionally, not geometrically. *FFGFT itself does not need this distinction*: it carries a single, curvature-faithful winding, and the sectors as well as the M_τ/M_B split serve here only as an *analysis* of that framework, not as native FFGFT structure. The present document therefore does not redefine these objects; it proposes a separate, FFGFT-side bridge reading. Primary source: Krüger, M. (2026), *From Physical State Evolution to Experienced Time*, Zenodo, doi:10.5281/zenodo.21302278. Documented lineage: Krüger, Feeney & Wende (2026), *Medinformatics*, doi:10.47852/bonviewMEDIN62029043; Krüger & Feeney (2026), *Smart Wearable Technology*, doi:10.47852/bonviewSWT62029484.

.2 U3 cannot be isolated

At one point the price is even an incoherence. Memory *is* a relation to time — a state that carries something about a before. Accumulation means integrating over a sequence. Without ordering (U1) there is no sequence to integrate over; without a present (U2) there is no now relative to which something is *past*. U3 is therefore not an independent fourth quantity beside U1 and U2, but what arises *when* ordering and present act together over a horizon. Marcel's own characterisation gives it away: "path-dependence, non-Markovian history" — "path" is ordering, "history" is past present. U3 is not even definable without U1 and U2. A test that lifts U3 out of time and probes it as pure storage over fixed episodes removes exactly the structure in which memory has meaning.

.3 The diagnosis: the memory reflex

The hierarchy "memory → experienced time" is thus wrong in its direction. It makes memory the generator of time, although memory presupposes time. This is structurally the same reflex Doc. 301 diagnosed for information: just as "it from bit" mistakes the ultimate reduction (the bit) for the first creation, "experienced-time-from-memory" mistakes the derived carrying form (memory) for the origin of time. The reversibility test (Doc. 301) reverses the order in both cases. For information: in the beginning was the geometry, and the word reads out one of its scales. For time: in the beginning was the unrolled winding, and memory is a particular way in which a system, on one of its scales, *carries* that winding. This is a cultural-cognitive hypothesis about *why* the intuition pulls (P35), not a mathematical statement; the sections that follow are.

.4 The FFGFT position: time is $D(k)$, fundamental

In FFGFT, time is the mass-circle, woven into the other circles through $\tilde{T} \cdot m = 1$ and not a separable factor (Doc. 285). The unrolled time-winding is the logarithmic spiral that the K_{frak} recursion forces (Doc. 295); its cumulated defect

$$D(k) = \sum_{n \leq k} 100 \xi_n \rightarrow \ln\left(1 + \frac{k}{75}\right) \quad (.1)$$

is the measurable time development (Doc. 296). Time is the recursion, not its origin: $D(k)$ has no privileged beginning, grows without bound, reaches no end — no first step, no last, no external standpoint ($\partial T^4 = \emptyset$, Doc. 301). Time is therefore fundamental — the third level, carried losslessly — not a product of memory. Memory cannot generate time, because time already *is* $D(k)$ before any system carries it.

.5 The decisive finding: the accumulation boundary lies in time

The remarkable point is that the distinction which recurs in cognition as M_τ versus M_B already stands, in the corpus, *at the level of time* — and is there already drawn against exactly the same bounded memory kernel. Doc. 295 contrasts FFGFT's *geometric accumulation* $D(k) = \sum 100 \xi_n \rightarrow \ln$ (unbounded, $1/n$ decay, ratio e) with a *bounded memory kernel*: if the defect per step stays constant, $D(k)$ grows linearly rather than logarithmically ($D(10^5) = 1333$ against $\ln(1 + k/75) = 7.2$), and the winding is Archimedean, not a log-spiral. This case of constant defect ($d = 1/75$ per turn fixed; the bounded, forgetting kernel) corresponds to the Markov fallback. The case $d = 0$ is different: without defect the winding closes after every turn, $D(k) \equiv 0$, nothing accumulates. (Corrected on 1 Oct. 2026: previously "The limiting case $d = 0$ (the bounded, forgetting kernel) corresponds to the Markov fallback" – d is the closure defect per turn (Doc. 295); $d = 0$ gives $D \equiv 0$, the linear growth $D(10^5) = 10^5/75 = 1333$ belongs to fixed $d = 1/75$.)

The mapping is then direct:

- M_τ — the finite, forgetting kernel — is the *Markov limit of constant defect* of $D(k)$ ($d = 1/75$ fixed): a bounded window, constant defect, linear rather than logarithmic growth. (Corrected on 1 Oct. 2026: previously "degenerate $d = 0$ / Markov limit" – see above; $d = 0$ would give no growth.)
- M_B — the accumulative, recursive structure — is a system that carries a $D(k)$ -like, unbounded integral of its own history.

The accumulation-versus-kernel boundary is therefore *not native to memory*; it is native to time. Memory that accumulates is memory that carries a $D(k)$ -like integral; a finite kernel is precisely the form of constant defect (Corrected on 1 Oct. 2026: previously "degenerate $d = 0$ form" – see above.) Hence — *on this reading* — M_B does not generate experienced time but appears as a cognitive-scale *instance* of the $D(k)$ accumulation that time *is* (as a bridge, not a derivation; see the limit of the bridge below).

Status of this identification. At the level of time the equation is corpus-anchored: Doc. 295 explicitly equates FFGFT's $d = 0$ closure with Marcel's Markov fallback. (Note of 1 Oct. 2026: elsewhere Doc. 295 itself describes the Markov case with constant defect and linear $D(k)$; M_τ fits that case, not $d = 0$.) The extension to the cognitive objects M_τ/M_B is, however, a further, FFGFT-side step — a *bridge* (Category B, a compatibility candidate), not a derivation. It is local to its assumptions and needs its own licence; anyone who reads the cognitive M_τ/M_B differently may decline the bridge without touching the time-level finding (the accumulation boundary in $D(k)$). The statement " M_B is an instance of $D(k)$ " is therefore *restricted*, not *proven* — a declared bridge candidate (Category B), not a proven identity. Naming it as a bridge rather than promoting it to a derivation is exactly the identity check Doc. 303 requires. This identity check, like the claim-state booking (restricted, Category B), belongs to the governance layer that goes back to Stefaan Vossen's *Dot Theory* and was co-developed in the IPI exchange (cf. Doc. 272); the identity-preservation record of Doc. 303 is the FFGFT-side extension on top of it.

.6 Consequence for the discriminator

This explains, from the FFGFT side, why the memory benchmark repeatedly tested the wrong half. A storage test over fixed episodes removes the temporal relation in which compression has any meaning; a test with exact, unbounded storage (full history) lets M_B coincide with lossless accumulation and makes its specificity invisible. The right touchstone is a *capacity-bounded, lossy* accumulator: a system that carries a *compressed* $D(k)$ -like integral — for that is exactly what a cognitive instance of $D(k)$ looks like. Yet even a passed test of that kind would show “semantic accumulation is distinguishable from generic compression”, a memory result — not “experienced time”. Experienced time is the $D(k)$ structure itself, which the test *presupposes* rather than produces.

.7 The measurement cut: the reference-less excerpt and the linearization reflex

The same inversion strikes measurement, not only memory. A measurement takes an *excerpt* of the time-winding. Even a lossless measurement — one that captures every step down to the quantum-bit limit (the winding quantum $\Delta w = 1$, Doc. 257/301) — delivers the *content* of the window in full, but not its *position* k on the spiral. Resolution (the content axis) and reference (the embedding axis) are different axes; losslessness saturates the first and leaves the second untouched. Synchronising successive measurements lengthens the excerpt, but it stays reference-less: one gains *relative* distances (clocked by the atomic clock), never the *absolute* position. The atomic clock is, by construction, a *uniform* ruler — constant pitch, hence exactly the M_τ limit of constant defect (Doc. 295) (*Corrected on 1 Oct. 2026: previously “ $d = 0$ / M_τ limit” – constant pitch means constant defect, not $d = 0$.*) —, and a uniform ruler laid on a non-uniform (fractal) structure cannot locate it. That there is no absolute time reference is thus boundarylessness at the level of measurement: no external standpoint ($\partial T^4 = \emptyset$) from which the position could be read.

The excerpt is nonetheless not empty. On the log-spiral there is no straight segment; every window carries its *curvature* — the local scale ρ , and with it the prior history — as folded-in form. The reference is not *lost* but *linearised*: the measurement reads the tangent instead of the arc — the local form of the same constant-defect approximation (*Corrected on 1 Oct. 2026: previously “ $d = 0$ approximation” – see above.*). Precisely, this is an error in the *accumulated precession angle* $\beta = 2\pi D(k)$ (Doc. 295): the uniform clock takes the local rate $d_k = 100 \xi_k$ as constant over the window ($\Delta\beta_{\text{unif}} = 2\pi w d_k$), whereas the true value is $\Delta\beta_{\text{true}} = 2\pi(D(k+w) - D(k))$, and the difference per 2π is not a new quantity but the local shape of the linear-versus-log gap of Doc. 295. For a window of w steps at position k , with relative window width $x = w \cdot 100 \xi_k = w/(k + 75)$:

$$e(k, w) = \frac{\Delta\beta_{\text{unif}} - \Delta\beta_{\text{true}}}{2\pi} = w d_k - (D(k+w) - D(k)) = x - \ln(1 + x). \quad (.2)$$

Three regimes follow; they are checked against the exact recursion $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ and are exact in the macroscopic limit, with an $O(1/k)$ correction at finite position (the closed form $D(k) = \ln(1 + k/75)$ carries a bounded, γ -like offset ≈ -0.0067):

- *Small window* ($x \ll 1$): $e \approx x^2/2 = \frac{1}{2}(100\xi_k w)^2$ (leading term of the closed form).
- *Scale-invariant window* ($x = 1$, a window one local scale wide): $e \rightarrow 1 - \ln 2 \approx 0.3069$, independent of k (asymptotically, $+O(1/k)$). This constant is the quantitative signature of the fractal self-similarity: the distortion of a “one-scale-wide” excerpt is the same everywhere on the spiral.
- *Macroscopic limit* (fixed physical w , growing k): $e \rightarrow 0$ as $\frac{1}{2}(w/(k + 75))^2$. Because we sit at large k (small ξ), the distortion is astronomically small — *this* is why measured time appears uniform and linear. Near the origin (small k , large ξ) it would be large; no measurement reaches there.

Hence the third reflex diagnosis (P35), twin of the word and memory reflexes: the apparent uniformity of measured time is a *linearisation artefact*, not a property of time. From an approximation in the measuring (tangent for arc) a property of the measured is falsely inferred (time *is* uniform). The excerpt carries its fractal history as curvature; the uniform clock erases it by construction; and because at our scale the error vanishes, we falsely conclude it was never there.

Booking. The error formula $e = x - \ln(1 + x)$ is the error in the accumulated precession angle $\beta = 2\pi D(k)$, verified against the exact recursion (script `ffgft_304_linearisierungsfehler_probe.py`); its three regimes are exact in the macroscopic limit (a consequence of $D(k)$, proven), with a named $O(1/k)$ correction at finite position. The exact coefficient of that correction (whether precisely C/k or carrying a log term) is obtainable from an Euler–Maclaurin expansion of $D(k)$ and is left open here — physically negligible at our scale, named for honesty. The equation “measurement = local linearisation of the excerpt” is a *declared assumption* about *this* kind of measurement (pointwise, tangential), not about every possible one — a measurement that *integrates* over a window would see part of the curvature. The reflex reading is philosophically speculative (P35).

.8 Congruence, self-similarity, and the survival of ordering

That there is *no reference* and that *all excerpts are equal* is one and the same fact: were there a distinguished excerpt, it would be a mark and hence a reference. But “equal” means different things in the two states of the time-winding, and the distinction is essential.

Rolled up (compact). With ξ_0 frozen the defect per step is constant, $d = 1/75$, the rotation number $74/75$ rational; the winding closes after *exactly* 75 turns and carries a 75-fold structure (Doc. 295). “Equal” here means *congruent*: 75 coincident segments, carried into one another by a $\mathbb{Z}/75$ rotation, exact and without remainder. A reference *does* exist here — but only *periodically*, modulo 75: one can name “part j of 75”, but not which turn, since all 75-cycles are congruent.

Unrolled (spiral). Letting the scale run ($\xi_{n+1} = \xi_n(1 - 100\xi_n)$) turns the rational 75-closure into the non-closing log-spiral of ratio e (Doc. 295, “unrolling”). “Equal”

now means only *self-similar*: the excerpts are *proportional*, not congruent — and only *asymptotically* so. This is exactly the origin of the $O(1/k)$ correction: the self-similarity is not a finite exact symmetry but a continuous one that sets in fully only in the limit. The periodic reference vanishes too, for the turns are no longer congruent but only proportional; the 75 marks smear into the continuous self-similarity.

Reference-lessness therefore holds *strictly only in the unrolled state*. Three reference states must be distinguished:

- *rolled up*: periodic reference (position modulo 75, exact);
- *unrolled, asymptotic*: no reference (pure proportion);
- *unrolled, finite* (near the origin): a weak absolute reference — the $O(1/k)$ deviation from self-similarity —, but one no measurement can reach.

What survives all of this is the *ordering*: which excerpt comes *before* which. It is a relation *between* excerpts, not an inner property of a single one, and is therefore untouched by the rescaling. It is U1, and it is the only invariant that survives the loss of absolute reference — exactly “time is the recursion, not its origin” (Doc. 301). What remains measurable of time thus falls into three levels: the *ordering* (exact, U1), the *relative interval* ($\Delta\beta = 2\pi\Delta D$, linearisation-distorted by $e = x - \ln(1+x)$), and the *absolute position* (absent, erased with the self-similarity). Ordering exact, interval distorted, position absent.

Refinement. “All excerpts equal” means *congruent* (rolled up, 75 parts, exact) or *self-similar up to scaling and rotation* (unrolled, asymptotic), not *identical*.

.9 What this means for the partition of U3

The measurement analysis feeds back onto the partition of U3. Since a memory is a system that carries a $D(k)$ -like excerpt (as shown in the decisive finding and the measurement cut), U3 inherits the same three-level structure: the *ordering* (which event before which) stays exact — it is U1 —, the *relative interval* (how long ago) is carriable but linearisation-distorted by $e = x - \ln(1+x)$, and the *absolute position* (the true place on the spiral) is absent, erased with the self-similarity. Crucially: *both* memory forms, M_τ and M_B , are reference-less — even a complete M_B knows only “so much accumulated D since an arbitrary start”, not “I am at turn k ”. The one thing M_τ too carries exactly is the ordering.

In the *FFGFT reading* the discriminating axis is not reach (finite window versus unbounded integral) but *curvature-fidelity* — offered as an *additional* candidate dimension for the bridge, not as a replacement of the operational M_τ/M_B definition (see the limit of the bridge below):

- M_τ presents a *linearised* excerpt of constant pitch — the limit of constant defect (*Corrected on 1 Oct. 2026: previously “ $d = 0$ / rolled-up limit” – see the section on the memory kernel.*). It reads the tangent, discards the curvature, “experiences” a uniform past. M_τ is the memory that commits the linearisation reflex.
- M_B presents the *curvature-faithful* excerpt — the log-spiral with the $1/n$ -decaying defect and ratio e . It carries the non-uniform texture of the past: near events dense and fast, far events wide and slow (Doc. 296).

Reach and curvature-fidelity are *two* axes, not one. A long but *linear* kernel (constant pitch, merely with a large window) would be reach-strong yet remain

M_τ -like, because it does not see the self-similarity; a short but *curvature-faithful* accumulator would already be M_B -like. The benchmark (v0.3–v0.5) presses both into the single axis of reach.

Hence the *positive* criterion for the discriminator, and it is not a new one: it is the e -criterion of Doc. 295 (a winding carries ratio e exactly when the defect per step decays as $1/n$; script, section [6]), applied to U3. What is to be tested is not “finite window versus accumulation” but *curvature-fidelity at equal reach*: does the accumulator carry the $1/n$ -decaying pitch, the ratio e , the self-similar texture? That is exactly what distinguishes M_B from any uniform kernel, and it is sharper than any reach boundary.

Booking. The three-level inheritance (ordering exact / interval distorted / position absent) and the mapping M_τ = linearised, M_B = curvature-faithful are consequences of $D(k)$ (derivable). The e -criterion as a discriminator test is corpus-anchored (Doc. 295). That Marcel's *cognitive* M_τ/M_B are exactly these objects remains the bridge candidate from the decisive finding (Category B, restricted) — now with a testable criterion. That remembered time *really* carries the spiral texture (the phenomenological non-uniformity of remembering) is interpretation (P35).

.10 Provenance and the limit of the bridge

So that the meshing above is not mistaken for absorption, let the limit of the bridge be drawn explicitly, in language that is unambiguous even without knowledge of how it arose.

The correspondence is a restricted, optional bridge candidate, not a secured mapping. Two axes must be kept apart. The e -criterion itself — a winding carries ratio e exactly when the defect decays as $1/n$ — is script-verified against the exact recursion and derivable from ξ (*proven*, Doc. 295). Only the *correspondence* between this curvature-fidelity and the operational M_τ/M_B distinction is unsecured: on the logical axis *restricted* (Category B), on the empirical axis with no benchmark performed — where such a benchmark would be an independent, jointly sealed comparison, not an FFGFT-internal calculation. It is not a derivation, validation, containment relation, or ontological classification of the Spiral-Time Memory Hypothesis by FFGFT, and its rejection leaves the internal claims of both frameworks unchanged.

An internal property of FFGFT can show that, under FFGFT assumptions, a non-linear, history-dependent accumulation structure $D(k)$ arises on the time level; that result stands on its own. It cannot, however, independently establish that a semantic, self-bound, affectively weighted, predictively reusable memory architecture follows from it. That extension is precisely the bridge candidate.

The operational specification of M_τ and M_B is untouched. In the Spiral-Time framework it is operational and functional, not geometrical: M_τ as finite, weighted, or compressible physical memory and path dependence; M_B as recursive access, semantic organisation, affective weighting, identity binding, persistent state carry, and predictive reuse. Curvature-fidelity is an *additional, FFGFT-side diagnostic candidate*, not the criterion that separates M_τ from M_B in that framework, and it replaces none of those functions. In both directions: a system may carry a $1/n$ texture while possessing none of those biological-cognitive functions; and a biologically

relevant memory architecture is not ruled out merely because it does not instantiate the specific *e*-criterion. The curvature test therefore remains a separate bridge audit, not a redefinition of the native objects.

The relationship is symmetric, the provenance separate. Neither framework is presented as containing, grounding, explaining, or subordinating the other. The M_τ/M_B distinction, the operational M_B architecture, and the audit sequence (v0.3–v0.5) originate in Marcel Krüger's Spiral-Time Memory work (Krüger 2026, doi:10.5281/zenodo.21302278); $D(k)$, the *e*-criterion, and the FFGFT interpretation of temporal curvature originate in Johann Pascher's FFGFT corpus. A later formal comparison or curvature-fidelity audit would be a separate, joint bridge contribution, preserving the independent authorship and conceptual identity of both frameworks.

.11 The past is always present: linearisation lies in the projection

An addition sharpens the measurement cut and the partition just given. The *past content* is *always* in the measurement result, independent of the window's shape, whether read linearly or not. For $D(k)$ is cumulative: every point of the spiral carries the summed integral of the whole prior history. Even a one-step window sits at this value and inherits it whole. A small window does not make it *absent*, only more weakly *resolved*; a long or more accurate window makes it sharper. "Present or not" is the wrong alternative — the past is always present, at variable sharpness.

What is decisive is therefore not how the window *looks* (linear or curved) but what the result is *applied* to. The measurement yields a value; the linearisation happens only when that value is mapped onto a *projection* — onto a uniform ruler (the atomic clock, the linear time axis). There, in the projection, the curvature is straightened. The distortion is not a property of the measurement but of the projection surface onto which the result is laid.

This is the projection structure of the corpus (Doc. 257/301): *measured* time is a *second-level* object — a projection of the $D(k)$ excerpt onto a uniform ruler. The past content is present as a *state* of the third level — the $D(k)$ value itself —; yet this value is a lossy résumé of the prior history, and the projection to the second level is lossy, and *that projection is the linearisation*. Not the measurement linearises, but the projection of the measurement result. Thus $e = x - \ln(1 + x)$ is, more precisely, the error of the *projection* onto the uniform ruler, not an error of the measurement itself.

Three things must be separated: the *presence* of the past as a state (always, third level: the $D(k)$ value is exact, but not the recovery of the prior history from it); the *sharpness* of its reconstruction (window- and accuracy-dependent); and the *linearisation error* (a property of the chosen projection, not of the measurement). The window's shape is not decisive for the *whether* of the past or for the *location* of the linearisation — but it is decisive for the *how sharply* of the reconstruction.

This sharpens the earlier talk of "carrying" the curvature: M_τ and M_B both *contain* the past — every $D(k)$ excerpt does —; they differ in the *projection*, not in the *content*. M_τ projects onto a uniform ruler (linearises, discards the curvature at projection), M_B projects curvature-faithfully (keeps the self-similar texture). The load-bearing axis

from the previous section — curvature-fidelity, not reach — is thus more precisely *projection-fidelity*: which projection the state passes through, not what it contains.

Booking. The cumulativity of $D(k)$ (the past always present) and the location of the linearisation in the projection (second level, Doc. 257/301) are derivable and corpus-anchored respectively; e as a projection error is the more precise form of the earlier formula. The separation presence / sharpness / linearisation is conceptual bookkeeping.

.12 Inverse inference and the holographic limit

The model is unrolled from the *measurement side*, for what stands at the centre is the application: one wants to infer from measurement results back to *causes*. This is the inverse direction of the reversibility test (Doc. 301) — not from cause to effect, but from the measured effect back to the cause. It is exactly this aim that makes the preceding analysis practically relevant: it asks *how far* a measurement result supports the inference to the whole.

And here is the limit, which was drawn too weakly before. From a small excerpt one may *not* infer the whole. The excerpt gives at best a *blurred* image of the whole — like a fragment of a holographic recording, which carries the entire image but only unsharply, at lower resolution. “The past is present” thus means, more precisely: it is folded in as a *blurred signature* — the cumulated value $D(k)$ is a sum from which the summands cannot be recovered —, not as a reconstructible recording. *Presence is not reconstructibility.*

The holographic structure is the self-similarity itself: because every excerpt is *similar* to the whole, it carries a scaled, blurred image of the total structure. From this follows something sharper than the earlier talk of “accuracy”: the sharpness of the reconstructed whole depends on the *size* of the excerpt (how much of the spiral), not on the fineness with which a small piece is read. A small window, however losslessly — down to the winding quantum — it is measured, gives a blurred whole; only more excerpt sharpens it. This is why the absolute reference is missing even at perfect resolution.

The impermissible step — from part to whole — is thereby nameable: an *extrapolation reflex*, taking the fragment for the image. It joins the family of the word, memory, and linearisation reflexes: in each, something derived, reduced, or partial is taken for the original, the whole.

For the inverse inference the exact reach follows: inferring from measurement results to causes is possible, but *only up to the holographic blur*. The rolled-up (compact, 75-exact) side — the underlying geometry, the “cause” — is *inferred*, not measured; the unrolled (spiral, measurement) side is where one stands. Rolled and unrolled are here two readings of the one winding; “cause” means the direction of the inference, not an ontological layer beneath the fundamental $D(k)$. Absolute position and ultimate origin are never sharply recoverable from an excerpt. This connects to the lossiness of accumulation: a $D(k)$ -carrying memory (M_B) is a *lossy* accumulator precisely for this reason — the holographic blur is the compression.

Booking. The measurement-side orientation and the inverse aim (measurement → cause) are methodological framing. The holographic limit on inverse inference follows from self-similarity and reference-lessness (derivable), as does presence $\neq$

reconstructibility. “Holographic” is here explicitly an *analogy*: whether every excerpt carries information about the *whole* (including the distant) — strong holography — or merely resembles its local neighbourhood — weak self-similarity — is an open question to be tested (P35), not an established theorem.

.13 Ordering of the levels

Three statements must be kept apart, each at its own temperature. *Mathematical*: U3 presupposes U1 and U2, and $D(k)$ is the accumulation structure derivable from the K_{frak} recursion (Doc. 295/296); and the linearisation error $e = x - \ln(1 + x)$ of a measurement window is verified against the recursion — consequences, no speculation marker. *Philosophical (P35)*: the memory reflex and the linearisation reflex (measured uniformity as an artefact) as cultural-cognitive inversions, counterparts of the word reflex (Doc. 301). *Posit*: that time is fundamental (the third level, $D(k)$ as something real that is carried) is the same realist posit as in Doc. 301, not a new claim. The deflationary reader may always say: all only description. The argument of this document is independent of that posit — read even as mere description, U3 remains incoherent without U1 and U2, and the accumulation boundary remains a property of the time structure, not of memory.

.14 Status

This document is a statement of position, not a new derivation: it brings together results from Doc. 285 (time as the woven mass-circle), Doc. 295 (log-spiral, $D(k)$, accumulation against a bounded kernel), Doc. 296 ($D(k)$ as measurable time, boundarylessness), and Doc. 301 (three levels, word reflex, self-projection) and Doc. 257 (the winding quantum as the bit limit) onto the question “memory or time first?”. The FFGFT-internal core of this document — the thesis that memory presupposes time, the linearisation reflex, and the measurement and projection epistemology — stands on its own; the M_T/M_B correspondence is the illustrative, *restricted* overlay from the bridge context and carries nothing the core needs. A measurement-theoretic extension places the reference-less excerpt, the congruence/self-similarity distinction (rolled up: 75 exact parts; unrolled: only asymptotically self-similar) and the linearisation reflex; the error quantity $e = x - \ln(1 + x)$, with the scale-invariant constant $1 - \ln 2$, is verified against the recursion. It is framed as a counter-position to a “memory → experienced time” hierarchy, in a neutral tone and without personal address; the M_T/M_B line is placed, factually, as a cognitive instance of the $D(k)$ structure (booked as a *bridge candidate, restricted*, Category B), not as its origin. The relationship to the Spiral-Time Memory Hypothesis is symmetric, the bridge optional and rejectable; provenance and limit are set out in a dedicated section. The mathematical statements are corpus-supported; the reflex diagnosis is philosophically speculative (P35) and booked as such.